Study 3b: clutter findings across models

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Overview

The main findings of Study 3 hold up in three different AI models, not just the one we started with. In all three, clutter that looks like the real thing ("lookalikes") wrecks counting: even Molmo, which counts clean pictures almost perfectly (97–99%), drops to 4–19% when lookalikes are added. All three also read words written in the picture and sometimes copy them into their answer, and all three follow lines better when the lines are coloured. Some predictions held only in some models, and one (more line crossings make it worse) held in none. Two more models, Gemma 4 and a bigger Qwen (8B), are still running on another account; their results will be added when they finish.

What we did

We showed each AI model the same 12,500 pictures and asked it to count shapes or objects, or to say where a line ends. Every model got exactly the same questions and was scored the same way.

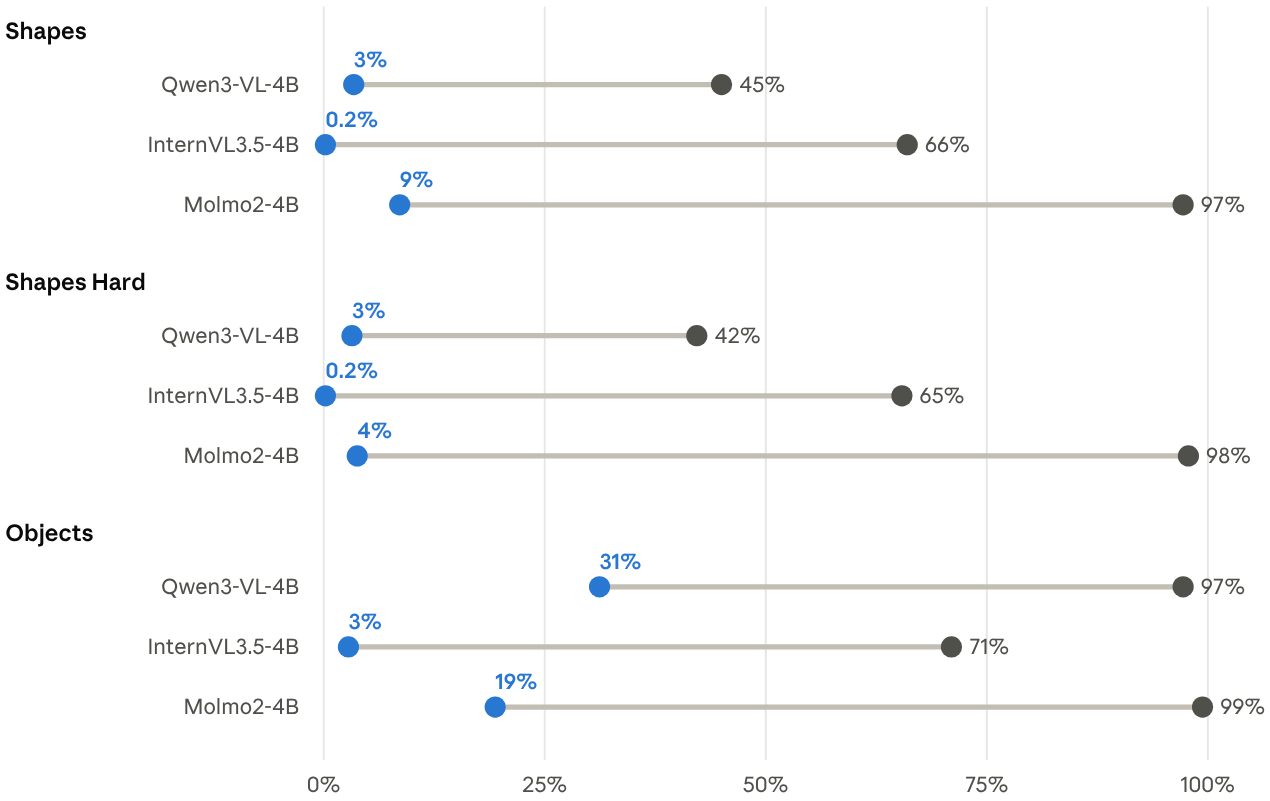
- **The pictures** come from Mohid's four datasets. Each scene appears once clean and several times with a different kind of clutter added: squiggles, blobs, lookalikes (almost-right shapes, like a circle with a gap), written words, or other lines.
- **The models** are all about the same size (4 billion parts), from four different companies: Qwen3-VL (Alibaba, our first run), InternVL3.5 (Shanghai AI Lab), Molmo2 (Allen AI) and Gemma 4 (Google, still running).
- **The rule we set before running:** a finding counts as general if it holds in at least 3 of the 4 models. Otherwise it belongs to one particular model.
- **Accuracy** means the model got every count in the picture right. "Held" means the effect was clearly bigger than zero, allowing for chance (a 95% range that doesn't include zero).

The headline: lookalikes beat every model

Adding lookalikes drops every model to between 0% and 31%, however well it counts clean pictures.

Lookalikes wreck counting in every model, even near-perfect Molmo

Share of pictures with every count right (%), 500 scenes each. Grey = clean picture, blue = same scene with lookalikes.



Kaggle runs on Zoya's and Maryam's accounts, scored by each dataset's own scorer · 3,000 pictures per model (clean and lookalike versions of 500 scenes in 3 datasets)

Molmo shows this best. It is the strongest counter on clean pictures, so its drop can only come from the lookalikes, not from the task being too hard.

Each prediction, and whether it held

Four predictions held in all three models so far, four held in some, and one held in none.

Prediction	In plain words	Qwen	InternVL	Molmo	So far
H1 Names and looks	Clutter the model can name (a heart) hurts more than clutter it can't (a squiggle); clutter that looks like the real thing hurts most	Held	Held	Held	General

Prediction	In plain words	Qwen	InternVL	Molmo	So far
H2 Busier squiggles	Squiggles with more curves hurt counting more	Held	Held	Held	General
H3 Every clutter hurts	Any added clutter makes counting worse	All 21 versions	20 of 21	19 of 21	Mostly
H4 Lookalikes overcount most	Lookalikes make the model count too many, more than any other clutter	Shapes only	All 3 datasets	Shapes only	Mixed
H5 Broken shapes counted as whole	The model counts an almost-right shape as a real one more than half the time	About half (50–52%)	Held (85–86%)	Held (53–70%)	Mostly
H6a Words hurt more	Written words hurt more than the same amount of other clutter	Objects only	Held	Held	Mostly
H6b Words get copied	When "circle" is written in the picture, the circle count goes up	Held	Held	Held	General
H7 Line crossings	More crossing lines make it harder to follow a line	Opposite	No effect	No effect	Not supported
H8 Colour helps	Coloured lines are easier to follow than black ones	Held	Held	Held	General

Lookalikes and broken shapes (H1, H4, H5). If a shape is close enough to a circle, the models call it a circle without checking that it is complete. InternVL does this 85% of the time. Stars fool every model most; squares fool them least.

Words in the picture (H6). The models don't only look at shapes, they also read. A word like "circle" printed in the picture raises the circle count in about 10–23% of cases, against under 2% without the word. Molmo resists words best: on Shapes Hard it stays at 78% accuracy with words added, against 20–28% for the other two.

Squiggles and names (H2). More curves hurt in every model. When we hold the number of curves fixed, squiggles the model can put a name to ("C", "S", "arrow") hurt more in InternVL and Molmo, which supports the original "names" idea. In Qwen there was no clear effect.

Lines (H7, H8). More crossings didn't make any model worse. The reason is that the models mostly guess that a line ends straight across from where it starts, rather than tracing it. Colour helps them trace: 12–20 points better than black lines.

What to be careful about

The results are strong but not final, mainly because two models are still running.

- **Two models still to come.** Gemma 4 and the bigger Qwen (8B) are still running. Under our rule, the four "General" findings already hold in 3 models, so they stay general whatever Gemma shows. The "Mostly" and "Mixed" rows can still change.
- **Some models are weak even without clutter.** Qwen gets only 42–45% of clean shape pictures right, and InternVL 65–71%. When a model already fails on clean pictures, the harder versions all sit near 0% and can't be told apart. Molmo (97–99% clean) is the fairest test.
- **Simple squiggles don't bother a strong model.** Molmo is as accurate with 2-curve squiggles as with no clutter. That is why H3 misses a few versions.
- **The names test is still rough.** The models give a name to almost every squiggle (InternVL names 98% of them), so "how nameable" barely changes between versions. The names result in H2 rests mostly on the simplest squiggles.
- **Lines tell us about guessing, not tracing.** Because the models mostly guess, the crossing test (H7) couldn't measure what we meant it to.